CALL FOR BSc AND MSc STUDENTS / 2026-2027

BSc and MSc Thesis Opportunities

HCAI Sarajevo invites students interested in computer vision, explainable AI, robotics and educational technology to get in touch about the following thesis topics.

APPLICATION DEADLINE

29 October 2026

Surgical robotics in a training environment

2 BSc and 2 MSc topics | MCI Innsbruck, Austria | pages 2-3

Collaborative robotics and textile manipulation

2 BSc and 2 MSc topics | Jožef Stefan Institute, Slovenia | pages 4-5

AI for education and programming

2 MSc topics | No CEEPUS mobility | page 6

Apply by email to Professor Senka Krivić

senka.krivic@etf.unsa.ba

Please mention the topic you are interested in and your study level.

The eight robotics topics include a planned CEEPUS research visit. The two education topics do not include CEEPUS mobility. The arrangements for each research visit are confirmed with the coordinator and host institution.

MCI INNSBRUCK / AUSTRIA

BSc topics: vision and robot monitoring

Both topics include a planned research visit to MCI Innsbruck through CEEPUS.

AT-B1 / BSc THESIS

Explainable Computer Vision for Surgical Robot Manipulation

This thesis explores computer vision methods for recognising objects, surgical tools and manipulation outcomes in a robotic training environment. It looks at how visual explanations can help users understand a model's predictions and spot errors.

You could work with images or video from manipulation tasks, test a recognition model and examine which parts of an image influence its predictions. The aim is to understand both when the model works and why it sometimes gets things wrong.

Possible outcome: A recognition prototype with visual explanations and an evaluation of its strengths and limitations.

AT-B2 / BSc THESIS

Visual Monitoring of Robot Arm and End-Effector Movements

This thesis investigates camera-based monitoring of a robot arm and its end-effector, the tool attached to the end of the arm. It focuses on detecting unexpected or potentially unsafe movements and giving users clear feedback to support responsible robot operation.

You could track movements in recorded or live video, define a small set of situations to monitor and design feedback that explains what has been detected. Evaluation would consider missed events as well as unnecessary warnings.

Possible outcome: A monitoring prototype and an assessment of how well it detects and communicates selected movement anomalies.

These topics concern robotic training environments, not clinical work with patients.

MCI INNSBRUCK / AUSTRIA

MSc topics: reliable and shared control

Both topics include a planned research visit to MCI Innsbruck through CEEPUS.

AT-M1 / MSc THESIS

Explainable and Uncertainty-Aware Robotic Manipulation

This thesis investigates how computer vision, uncertainty estimation and explainable decision-making can support reliable robotic manipulation on surgical training platforms such as the mini da Vinci. It asks how a robot can explain its actions, recognise its limitations and decide when human intervention is needed.

You could connect visual estimates to manipulation decisions, compare ways of deciding when to ask for help and test whether the system's explanations reflect the information behind its actions.

Possible outcome: An experimental system that combines action explanations with uncertainty estimates, evaluated across selected manipulation scenarios.

AT-M2 / MSc THESIS

Explainable Shared Control for Surgical Robotics

This thesis explores explainable AI for collaboration between a human operator and a surgical robot, such as the mini da Vinci, in a training environment. It examines how the robot can explain its assistance or interventions and suggest alternative actions that support human oversight and appropriate trust.

You could develop a shared-control scenario in which the operator remains involved in the task, then compare ways of explaining the robot's assistance. A key question is whether users understand when and why the robot intervenes.

Possible outcome: A shared-control prototype and an evaluation of the clarity and usefulness of its explanations.

These topics concern robotic training environments, not clinical work with patients.

JOŽEF STEFAN INSTITUTE / LJUBLJANA, SLOVENIA

BSc topics: understanding robot behaviour

Both topics include a planned research visit to the Jožef Stefan Institute in Ljubljana through CEEPUS.

SI-B1 / BSc THESIS

Visual-Language Models for Understanding Robotic Textile Manipulation

This thesis explores the use of visual-language models to recognise textile states and describe human and robot actions during manipulation. It evaluates how accurately these models interpret visual observations and where they fall short in collaborative scenarios.

You could build a small collection of task examples, compare model descriptions with reference observations and study errors such as missed actions or details that are not actually visible. Visual-language models combine information from images and language.

Possible outcome: An evaluation of selected models, supported by examples that show what they can and cannot reliably describe.

SI-B2 / BSc THESIS

Visualising Robot Intentions in Human-Robot Collaboration

This thesis investigates how visual feedback can communicate a robot's planned movements, task progress and safety-related warnings during collaborative manipulation. The goal is to develop and evaluate an interface that helps people understand and anticipate the robot's behaviour.

You could design a display of the next planned action, the current task stage and relevant warnings, then test whether people interpret those messages as intended.

Possible outcome: A working interface and an evaluation of how clearly it communicates the robot's intentions.

The exact scope, equipment and evaluation plan will be agreed with the supervisor.

JOŽEF STEFAN INSTITUTE / LJUBLJANA, SLOVENIA

MSc topics: planning and decision-making

Both topics include a planned research visit to the Jožef Stefan Institute in Ljubljana through CEEPUS.

SI-M1 / MSc THESIS

Explainable and Safety-Oriented Motion Planning for Human-Robot Collaboration

This thesis investigates motion planning that takes human presence and safety constraints into account during collaborative textile manipulation or assembly. It explores how a robot can explain its chosen movements, alternative actions and reasons for stopping or replanning.

You could compare planned movements in defined collaboration scenarios and develop explanations for route choices and changes of plan. Evaluation would examine the behaviour of the planner alongside the explanations it gives.

Possible outcome: A motion-planning prototype with explanations and an experimental analysis of its behaviour under selected constraints.

SI-M2 / MSc THESIS

Explainable Multimodal Decision-Making for Deformable Object Manipulation

This thesis investigates how visual-language models and symbolic task planning can be combined for robotic manipulation of deformable objects, particularly textiles. Decisions and explanations should be grounded in observed object states, task constraints and perception uncertainty. Both task performance and explanation faithfulness are evaluated.

You could connect visual observations to a task planner and trace the information used to choose an action. The central question is whether an explanation reflects the system’s actual decision process, rather than simply sounding plausible.

Possible outcome: A multimodal prototype and an evaluation of task performance and the faithfulness of its explanations.

The exact scope, equipment and evaluation plan will be agreed with the supervisor.

HCAI SARAJEVO / EDUCATIONAL TECHNOLOGY

MSc topics: AI in education

> **NO CEEPUS MOBILITY**
> Neither of the two topics below includes a CEEPUS research visit.

ED-M3 / MSc THESIS / NO CEEPUS MOBILITY

LLM-Assisted Educational Resource Authoring and Learning Analytics

This thesis investigates how large language models can help teachers create educational quizzes through a web-based platform with human review. The platform would support generating, editing and delivering quizzes, while collecting interaction data for future learning analytics and educational research.

You would develop the workflow from a model's draft questions to a teacher's review and a learner's attempt, while considering question quality and learners' privacy.

Possible outcome: A web platform for quiz authoring and delivery, with a data model and an initial evaluation of the generated material.

ED-M4 / MSc THESIS / NO CEEPUS MOBILITY

Human-in-the-Loop Generation of Alternative Programming Solutions

This thesis focuses on a system that uses large language models to generate several valid solutions to a programming task, with a person involved in reviewing the results. The aim is to make it easier to build a diverse collection of correct solutions for personalised programming education.

You would build a process for generating, testing and reviewing solutions, checking both correctness and whether they represent different approaches to the task.

Possible outcome: A prototype and an evaluated collection of alternative solutions, with a record of how they were checked.

Human review is central to both projects, alongside testing and responsible use of learner data.

RESEARCH VISITS AND APPLICATIONS

Working on your thesis abroad

The eight robotics topics are linked to planned CEEPUS visits: four at MCI Innsbruck in Austria and four at the Jožef Stefan Institute in Ljubljana, Slovenia. The visits are intended for thesis-related research, laboratory work and discussions with the host team.

Who can take part?

CEEPUS supports bachelor's and master's thesis visits. General eligibility includes two completed semesters and citizenship of a member country or an accepted equivalent status. The coordinator will confirm how these conditions apply to your study cycle and proposed visit. [1]

How long would the visit be?

Austria's OeAD lists short thesis visits of 1-2 months. The Slovenian information portal lists a one-month short stay, with at least 21 days in the host country. The actual dates and length of these visits will be agreed with the host. [2, 3]

What about funding?

CEEPUS support is awarded by the host country. The grant and arrangements for accommodation and insurance depend on the destination and current rules. Any travel reimbursement should be checked before booking. Choosing a thesis topic does not, by itself, confirm a scholarship. [2-4]

Apply by 29 October 2026

Email Professor Senka Krivić at
senka.krivic@etf.unsa.ba.
Mention your preferred topic and study level.

Further information

[1] CEEPUS: eligibility and applications

[2] OeAD: CEEPUS in Austria

[3] Study in Slovenia: exchange programmes

[4] HCAI Sarajevo: mobility information and local contact

Programme information checked on 25 October 2026. Research tasks and possible outcomes in this booklet indicate the direction of each topic; the final scope will be agreed with the supervisor.